COMPANY PROFILE · 2026

Vital Agri Nutrients

Crop nutrition engineered, manufactured, registered and tested in Pakistan.

Lahore · since 2010 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd

WHO VAN IS

We formulate, manufacture, register and test what we sell. Our only plant is in Lahore. Most of what we make is sold under our partners' brands.

23

registered brands, 22 of them made in our plant; any of those 22 can also be made under a partner's brand

34

licences from the Pakistan Standards and Quality Control Authority (PSQCA): 22 on our brands, 12 on partner-ready products

28

free crop plans, each giving the rates per acre for every crop stage

27

formulations in our library with no brand yet; 10 are finished and ready for a partner's brand

100+

brands of 14 partners since 2010, registered with VAN as the manufacturer

LAB 336

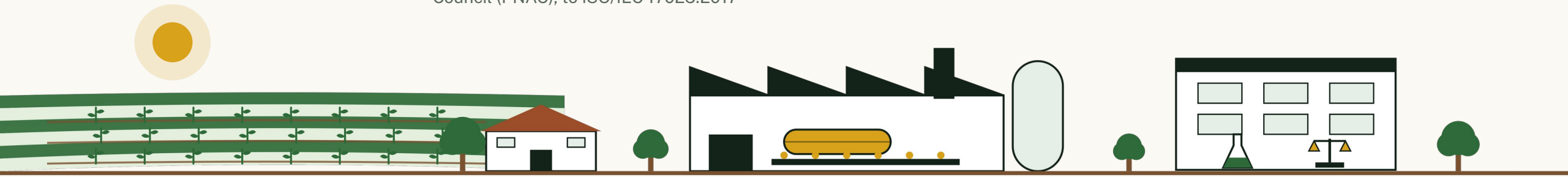
our laboratory's accreditation number from the Pakistan National Accreditation Council (PNAC), to ISO/IEC 17025:2017

144684

Pakistan Patent number for Vital Urea, our sulfur coated urea

150+

people work at VAN (2026)



OUR HISTORY

How we built VAN, 2009 to 2026.

2009 — First research farm

Acquired in 2008. Trials have run on it every season since 2009.

2010 — Company and land

The company was incorporated, and we bought the land next to the research farm to build the plant.

2011 — The plant

We started building the plant and made our first blended fertiliser.

2012 — Humic acid

We became the largest producer of humic acid in Pakistan.

2014 — The coating line

We added the coating line; its first product was boron-coated nitro potash, now sold as Vital Potash. We began soil analysis and a soil baseline study.

2015 — Crop and soil

We based our crop plans on the needs of each crop and each type of soil.

2017 — Our own shops

We opened the Vital Agri Centers.

2019 — Vital Urea

We launched Vital Urea, our sulfur coated urea. Vital Green was founded to sell our range direct to growers.

2023 — Potassium from ash

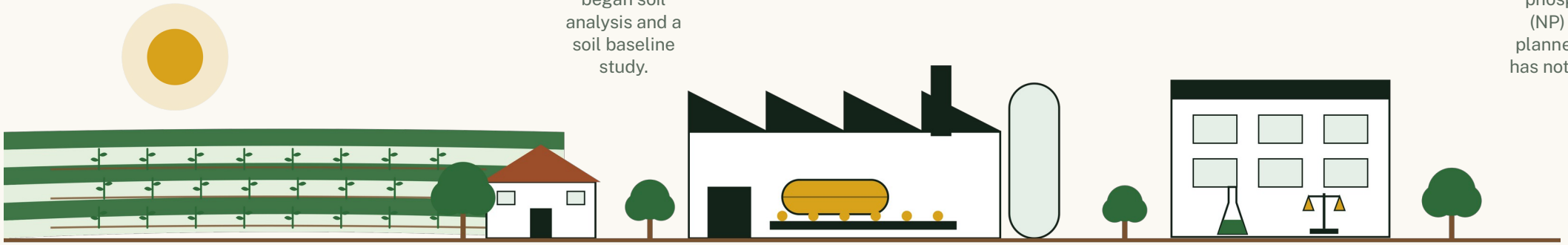
We made our first commercial batch of potassium recovered from crop residue ash.

2024 — Lab accredited

Our laboratory was accredited by PNAC to ISO/IEC 17025:2017 (LAB 336).

2026 — Patent and bio lab

Pakistan Patent 144684 was granted for Vital Urea. We opened our biological laboratory. A nitrogen-phosphorus (NP) line is planned; work has not started.



THE PLANT

5 production lines and 2 laboratories, on 1 site.

1 Granular

This line blends single-nutrient and speciality granules to a target ratio, after we check that the materials are compatible.

2 Coated

This line, added in 2014, applies the sulfur coating that makes Vital Urea, at commercial scale, and coats boron and iron onto granules.

3 Liquid

This line makes our VL range of liquid fertilisers, from foliar concentrates to grades for fertigation.

4 Water-soluble

This line makes fully water-soluble grades for fertigation and foliar feeding.

5 Compounded

This line makes granules that each contain all the declared nutrients in their declared proportions, so the nutrients stay together from the bag to the field.

• Biological laboratory

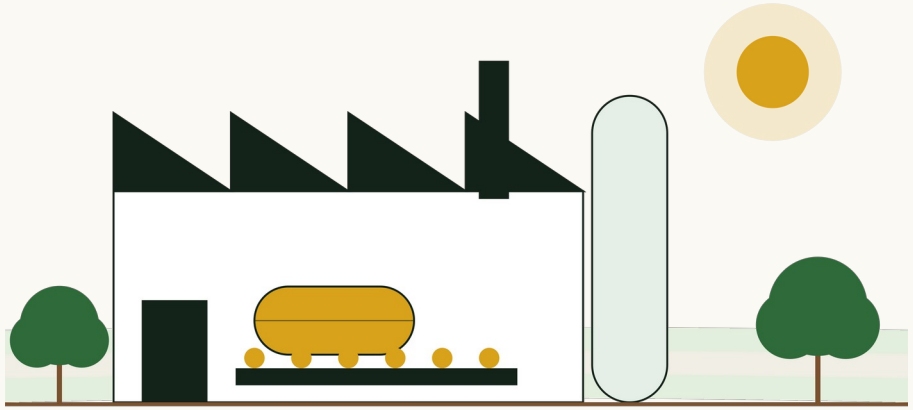
Opened in 2026. Locally isolated strains of phosphate-solubilising bacteria are in field trials.

• Analytical laboratory

Accredited by PNAC to ISO/IEC 17025:2017 (LAB 336).

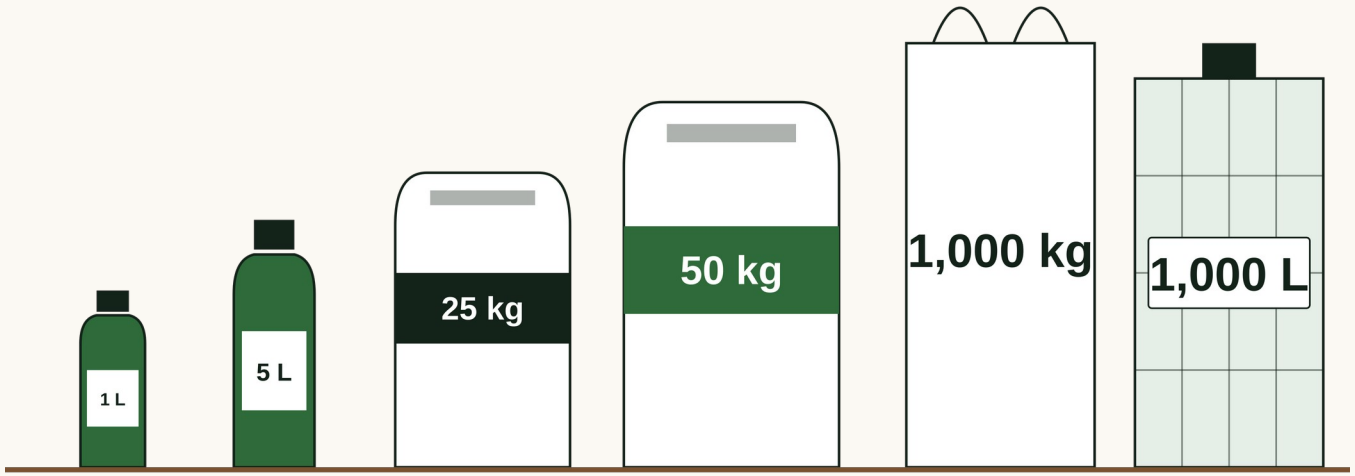
• Packs and storage

Packs range from a 1 L bottle to a 1,000 kg bulk bag, and finished goods are stored on site.



The analytical laboratory is on the same site as the 5 lines. Every batch from every line is tested there before it ships.

We can make the same composition in more than 1 physical form.



Physical forms

We make granules, powder, soluble crystals, liquid, pellets, water-dispersible granules and coated prill.

Pack range

Packs range from a 1 L bottle to a 1,000 kg bulk bag and a 1,000 L intermediate bulk container (IBC). Estate and contract-farming packs use our retail packing line.

1 t
the size of a production batch; for an order smaller than 1 t, we make a 1 t batch and pack the order from it

7
physical forms

1,000 L
the largest liquid pack, an IBC

None
there is no minimum order size

DOOR 1 OF 4

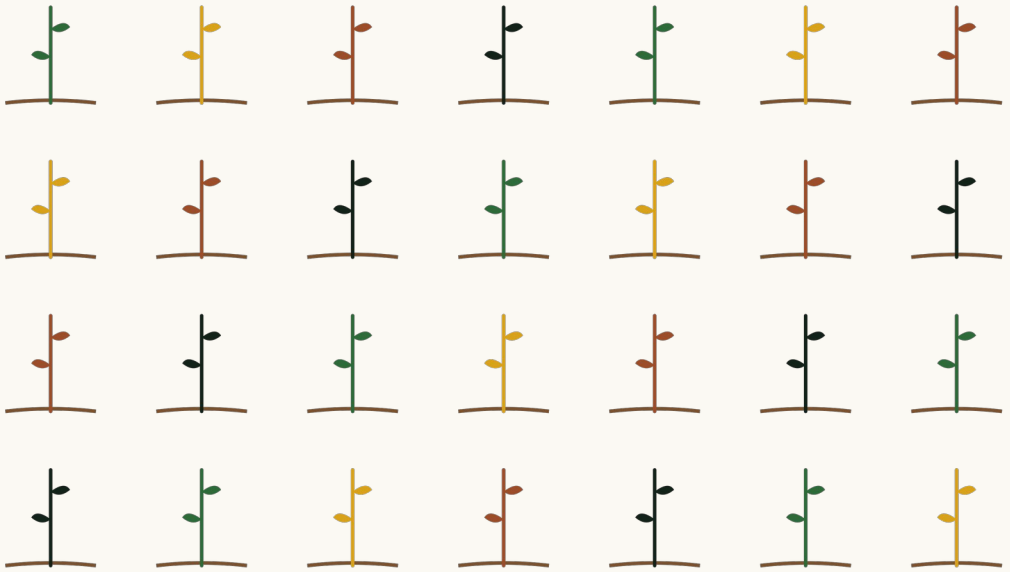
I grow



We publish 28 crop plans, each giving the amount to apply per acre at every crop stage. Your sowing date sets the dates of the stages, and 24 of the plans can also be adjusted to your soil. The crop plans, the Soil Atlas and the tools on van.com.pk are free, and the source of every figure is given.

- Crop plans van.com.pk/crops
- Soil Atlas van.com.pk/soil.html
- The 17 nutrients a crop needs van.com.pk/knowledge/nutrients.html

We also publish free tools to use with the 28 crop plans.



Wheat · Maize · Basmati rice · Hybrid rice · Cotton · Sugarcane, February planting · Ratoon sugarcane · Canola · Sesame · Soybean · Sunflower · Chickpea · Lentil · Mungbean and mash · Potato · Tomato · Chilli · Garlic · Watermelon · Turmeric · Onion · Banana, year 1 · Banana, year 2 · Citrus · Date palm · Mango · Strawberry · Guava

Vitalytics, the tools

On 24 crop pages, the Nutrition creator scales the crop plan to the acres you enter and adjusts the amounts to your soil report or your district.

Soil Atlas

It shows the results of 770,160 Punjab soil samples, tested in about 2016 to 2018, district by district, beside the nutrients taken out in the harvested crop.

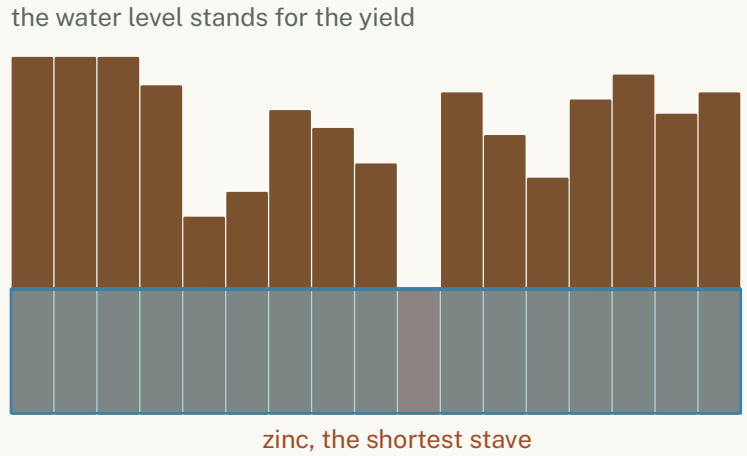
Yield and Discipline Simulator, beta

Yield depends on doing every task on time: sowing date, seed and plant population, fertiliser at the right stage, irrigation on schedule, and control of weeds, pests and disease. Crop nutrition is 1 of these tasks. The simulator scores these tasks for wheat and potato. We plan to add the other 26 crop plans; no date has been set.

Every crop plan is based on 17 years of trial records from the research farms we now run, starting in 2009.

A crop's yield is limited by the nutrient in shortest supply.

The rule is Liebig's Law of the Minimum, from the 1840s. A nutrient is short when the crop needs more of it than it can get, and adding more of the other nutrients then does not raise the yield. The drawing shows the law as a barrel: each stave is 1 nutrient, and water fills the barrel only to the top of the shortest stave. If nitrogen is not the shortest stave, another bag of urea only makes the nitrogen stave taller; the water still runs out over the shortest one.



3 from air and water

Carbon, hydrogen and oxygen are not bought as fertiliser. Together they make up more than 90% of a plant's dry weight.

3 primary nutrients

Nitrogen, phosphorus and potassium are needed in the largest amounts. Farmers spend most of their fertiliser money on them, and most problems come from too much of 1 and too little of the other 2.

3 secondary nutrients

A crop needs kilograms of calcium, magnesium and sulfur per acre, about as much of each as of phosphorus. Yet in Pakistan these are the nutrients most often left out of fertiliser programmes.

8 micronutrients

A crop needs iron, manganese, zinc, boron, copper, molybdenum, chlorine and nickel in grams per acre, but it cannot complete its life cycle if any of them is missing.

On van.com.pk the barrel is interactive. A grower chooses a crop and enters the bags applied to 1 acre. Each stave then shows how much of its nutrient those bags supply, as a share of the amount in our published crop plan for that crop. The shortest stave is the nutrient furthest below the crop plan. We make products that supply 11 of the 17 nutrients.

Pakistan applies more fertiliser nutrient per acre than the world average, and since 1999-2000 its wheat yield per acre has risen more slowly than its nutrient use per acre.

63 kg

of fertiliser nutrients (N, P_2O_5 and K_2O combined) per acre of Pakistan's cropland in 2023 (calendar year). The world average was 47 kg.

83 : 1

tonnes of nitrogen (N) per tonne of potash (K_2O) applied in Pakistan in 2023. The world applied 34 kg of potash for every 100 kg of nitrogen.

29%

nitrogen use efficiency in 2023 (FAO estimate): the share of nitrogen added to cropland that is taken out in the harvested crop. United States about 71%; world 55%.

+68% • +31%

nutrient applied per acre of arable land rose by 68% from fiscal year 1999-2000 to 2023-24 (July to June), and wheat yield per acre by 31% over the same years.

FAO estimates that, of every 100 kg of nitrogen added to Pakistan's cropland, about 26 kg are lost to the air and 11 kg are lost to water. Very little sulfur, magnesium, zinc or boron is applied in Pakistan, according to the national fertiliser figures.

Sources: Our World in Data / FAO, fertilizer use per hectare, 2023 (156 and 117 kg/ha, divided by 2.471). FAOSTAT, Fertilizers by Nutrient, 2023 (Pakistan N 3,808,775 t, K_2O 45,735 t). FAOSTAT, Cropland Nutrient Balance, December 2025 release (2023, FAO estimates). World Bank AG.CON.FERT.ZS and the Pakistan Economic Survey Table 2.5 (1999-2000 to 2023-24). All are charted and cited on the Knowledge pages of van.com.pk.

Most of the 770,160 Punjab soil samples are alkaline and below the adequate level of phosphorus.

8.22

mean soil pH; 96.1% of samples are above pH 7.5

0.61%

mean organic matter; 97.6% of samples are under 1%

97.9%

of samples are below the adequate level of phosphorus

87.8%

are below the adequate level of potassium

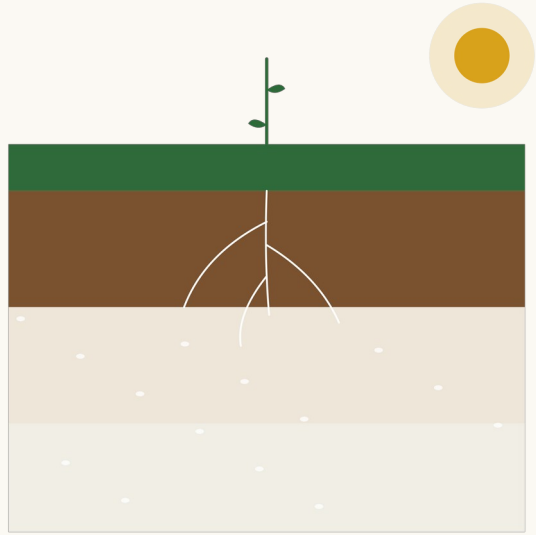
59.0%

are below the adequate level of zinc

51.7%

are below the adequate level of boron

0 survey samples were tested for sulfur in any of the 36 districts.



These data come from the Punjab provincial soil survey of 36 districts, whose samples were tested in about 2016 to 2018. We did not carry out the survey. We use it together with our own soil records: 18,000 from Sindh and 600 from Gilgit-Baltistan.

DOOR 2 OF 4

I sell



As a dealer, you are supplied direct from our plant with all 23 of our brands. We test every batch before it ships. A farmer can read the certificate of analysis for any bag on van.com.pk by entering the batch number printed on it. Our dealer terms are published on van.com.pk, and territory and margin are agreed with each dealer.

Become a dealer van.com.pk/become-a-dealer.html

Where to buy van.com.pk/where-to-buy.html

What we provide to dealers, and our terms.

The range

All 23 brands, in the pack sizes the trade sells, shipped within the times shown below.

28 crop plans

Your customers get a full crop plan to follow, stage by stage.

Brand material

VAN-branded literature and point-of-sale material.

Training and plots

Training for your team, and demonstration plots in your territory before a product is launched there.

1 t

a production batch; there is no minimum order. For an order smaller than 1 t, we make a 1 t batch and pack your order from it.

3 to 5

working days, usually, from order confirmation to shipment for 1 to 10 t. Orders of 11 to 100 t usually ship in 7 to 18 working days.

15

working days at most, from account opening to delivery of the first consignment

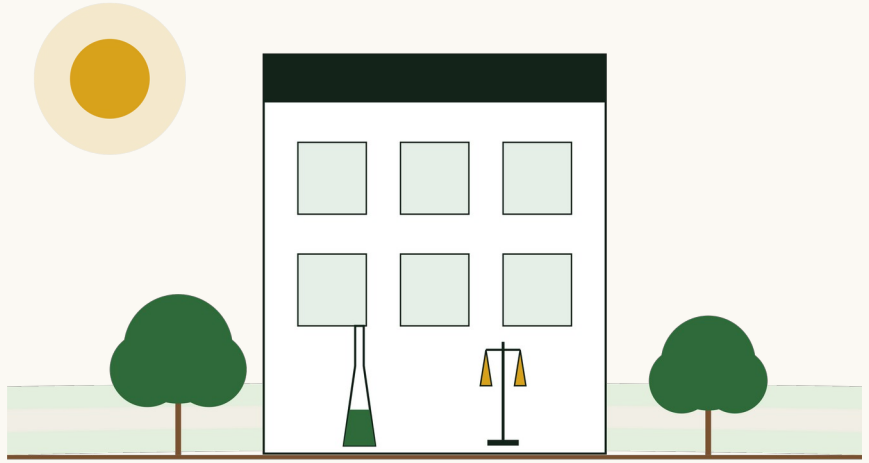
No credit

is given; we offer a cash discount for advance booking instead

Each dealer's territory and margin are agreed with that dealer, based on the products the dealer will carry and the volume the dealer will buy.

I SELL · QUALITY

We test every batch before it ships.



LAB 336

PNAC accredited, ISO/IEC 17025:2017, since 2024

8 of 21

laboratory tests are within the scope of our accreditation; the prices of all 21 are published

2

analysts test every sample, working independently

3 to 5

working days from confirmed booking to report, for samples customers send us

- A batch outside our internal limits, which are tighter than the product specification, is rejected.
- We use written test methods and calibrated instruments, and we analyse reference standards with every set of samples, so a result does not depend on who ran the test or on which day.
- A farmer can read the certificate of analysis for any bag on van.com.pk by entering the batch number printed on it.
- Our quality management follows ISO 9001, but certification by System Certification Centre (SCC) is still in progress, expected in 2027.

I SELL · REGULATORY

The PSQCA licence of each of our licensed brands is listed on van.com.pk with its licence number.

34

PSQCA licences: 22 cover 20 of our 23 brands, and 12 cover products ready for partners

8

Pakistan Standards under which the 22 brand licences are held

100+

partner brands registered with VAN named as the manufacturer

What each listing shows

Each listing gives the brand, its licence number and the Pakistan Standard under which it is held.

We handle the registration

For a partner, we file the registration under the partner's brand, with VAN named as the manufacturer. We have filed more than 100 of these registrations.

Label claims are checked before printing

We check every claim on a label before it is printed, and we will not print a claim that a trial does not support.

DOOR 3 OF 4

Made under my name



At our plant we have made products for more than 100 brands owned by 14 partner companies. You market and sell the product. We can take it from formulation to the finished bag, and file the registration under your brand with VAN named as the manufacturer.

- Your Brand van.com.pk/partner
- The 3 partnership models van.com.pk/partner#models
- How we build a range van.com.pk/partner#range

3 ways to work with us over the long term.

Each project follows 1 of these 3 models, and a partner often uses 2 models at the same time, on different projects. Commercial terms are agreed for each project and are not published.

Model	What we provide	What you provide	Registered under	Time to market
White labelling	We take a formulation we have already developed, trialled and registered, and make, test and pack it under your brand. We review the label and its claims before printing. Repeat lots are made to a fixed specification.	Your brand, your artwork, your market and the selling.	We file it under your brand, with VAN named as the manufacturer.	About 3 months to the first commercial batch, registration included.
Joint product development	6 stages from your brief: feasibility, bench formulation in our laboratory, a pilot on the commercial line, trials on our research farms and on your plots if you want them, the dossier and the registration.	The brief: market, crop, target price and the constraint the product must overcome. Your own demonstration plots, if you want them.	We file it under your brand, with VAN named as the manufacturer.	Most projects take 7 to 12 months. Across all 14 partners, projects have taken from 2 to 17 months.
Toll manufacturing	Your own formulation, made on our 5 production lines to your specification. Every batch is tested in our accredited laboratory (LAB 336) and released against a certificate of analysis.	The formulation, the specification, the brand and the market.	Agreed for each project: under your brand or under VAN's name.	Depends on the formulation and its registration. Agreed before the first lot.

Under every model, our own formulations and processes remain VAN's property.

We design a full range for your brand.

If your brand has no range of its own, you do not have to choose products from a list. We design the range around your crops, regions and sales channel, starting from products we already make, and we keep adding to it.

1 — YOU

Your crops, regions and channel

Tell us which crops your customers grow, in which districts, and whether you sell through dealers, to estates or to corporate farms. That is the complete brief for a first range.

2 — VAN

A launch range from products we already make

We propose the products a first season needs, each with its application stage and pack size, based on our published crop plans.

3 — VAN

A roadmap from the library

We plan what to add in the second and third seasons from the 27 formulations in the library, and say which would need a trial first.

4 — BOTH

Each product on its own timeline

A library formulation takes about 3 months to its first batch. A product developed to your brief usually takes 7 to 12 months. Products are added to the range 1 at a time.

Our formulation library holds 27 formulations. 10 are finished and ready to be sold under your brand, and the others are at earlier stages of development.

Our partner portfolio in figures.

A company that signs a non-disclosure agreement (NDA) with us can be given the details of this portfolio in a meeting.

100+

partner brands registered with VAN named as the manufacturer

14

partner companies own the 100+ brands

2010

the year our longest-running partnership began

27

formulations in our library with no brand yet; 10 are ready for your brand

2 to 17

months from brief to market: the fastest and the slowest so far

7 to 12

months for most projects developed to a brief

Our partners include multinational crop-protection companies, national agri-input companies, sugar mills and corporate farms, and importers who want a locally made product in place of one they import.

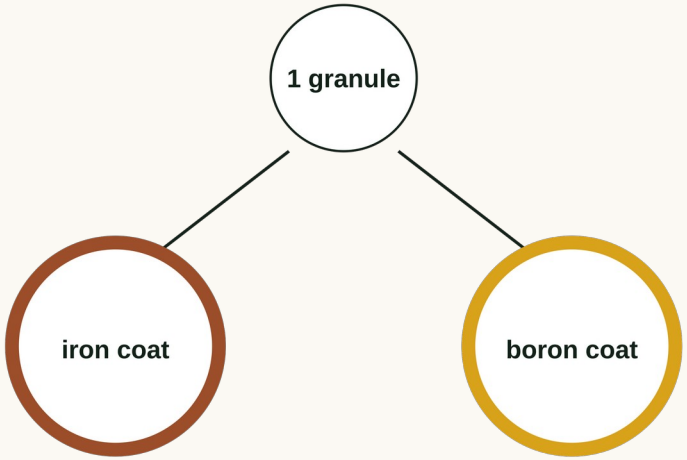
We design each grade for the soil where it will be used.

770,160
Punjab soil survey samples

18,000
our own soil records from Sindh

600
our own soil records from Gilgit-Baltistan

For example, we coat the same base phosphate granule with iron for bags sent to Kamalia and with boron for bags sent to Lodhran. Both are made from 1 base material on 1 coating line; only the micronutrient changes, to suit the soil where the bag will be used.



We do not sell your product for you

We provide no sales force, no dealers and no farmer demand. You own the market and the selling.

We do not hand over our formulations

Under every model, our own formulations and processes remain VAN's property.

No unsupported label claims

We will not print a claim that a trial does not support, even if your marketing department asks for it.

DOOR 4 OF 4

I want to do it differently



With our free tools, Vitalytics, a grower can enter a crop, acres, sowing date or soil reading and get an answer for that field. A company that wants a product that is not yet on the market can develop it with us. Each of our 3 flagship products was developed in response to a question that farmers ask.

- Vitalytics, the tools van.com.pk/tools.html
- The formulation library van.com.pk/partner/pipeline.html
- The composition chart van.com.pk/composition.html

THE 3 FLAGSHIPS

Our 3 flagship products address nitrogen loss, phosphate bound by calcium, and poor filling of grain, tubers and fruit.

Vital Urea · SCU

"I put 2 bags of urea on and the field goes yellow again in 3 weeks. Where did it go?"

Urea nitrogen is taken up by the crop or lost to the air or to water. The sulfur coating lets water reach only part of the urea at a time, so nitrogen is still available when the crop needs it.

Green Phosphate · 6-32-0

"A bag of DAP every year and the soil test still says phosphorus is low. Where does it go?"

In calcareous soil above pH 8, calcium turns phosphate into forms the root cannot use. Each humic-sealed granule creates an acidic zone that slows the reaction until a root reaches the granule.

Vital Potash · 11-0-44

"I fed the crop well and it still filled light. Why?"

Nitrogen is needed mainly for leaf growth, and potassium for filling grain, tubers and fruit. Vital Potash is made for the filling stage, with boron in 2 forms coated onto the granule since 2014.



FLAGSHIP 1 OF 3 · VITAL UREA

Vital Urea is a patented sulfur coated urea made in Pakistan. In the 2023 and 2025 trials, 1 bag of Vital Urea (25 kg) took the place of 1 bag of urea (50 kg) at the first nitrogen application.

144684

Pakistan Patent, held by VAN. Vital Urea was launched in 2019.

25 kg

bag. N 32%, S 13%: at least 8.0 kg of nitrogen and 3.25 kg of sulfur in every bag.

Maize trial	Year	Run by
Rafhan Maize Products	2023	Joint, VAN with Rafhan
Descon Research Farm	2024	Joint, VAN with Descon
A partner company's commercial trial, Sheikhupura	2025	The partner company

Some crop plans use Vital Urea at more than 1 stage; follow your crop plan. Our commercial team holds the trial summaries.

2012

We began work on coated products and have tested many combinations since.

Drill

at sowing, or side-dress it; broadcast if no drill. Do not apply it through fertigation.

In stages

once wet, the coating breaks down over days, not all at once.

We do not claim a yield figure from the 3 maize trials or from any other trial, or a laboratory release curve in days. We also do not claim that the coating stops volatilisation, or that Vital Urea changes soil pH or soil test results. We recommend comparing nitrogen products by the nitrogen that reaches the crop in the weeks it can use it, not by the weight of product.

How each product works, and what we do not claim.

Green Phosphate · 6-32-0 · 35 and 50 kg

Calcareous soil makes phosphate unavailable

Above pH 8, in soil with free lime, phosphate dissolving from a granule reacts with calcium within a few centimetres and comes out of solution; over time it changes into forms the root cannot use. In the Punjab soil survey (samples tested in about 2016 to 2018), 35 of 36 districts were rated Critical or Weak for phosphorus.

Each granule creates its own acidic zone

The granule is made in an acidic mix and sealed with humic, with a fraction of fulvic. The soil next to the granule is acidic at first, which slows the reaction between phosphate and calcium for the weeks a root needs to reach the granule. This effect is confined to a small zone around the granule (a microsite), so it does not change soil test results.

What the humic coating does

Humic substances benefit the soil and the root. A review of 81 published studies found that, on average, they increased shoot and root growth (Rose et al. 2014).

We do not claim that 70 to 80% of applied phosphate is fixed, that Green Phosphate outperforms diammonium phosphate (DAP) or triple superphosphate (TSP) as a phosphate source on its own, or that it changes a soil test.

Vital Potash · 11-0-44 · 20 kg

Why potassium matters at filling

In general, potassium moves the sugar made in the leaf into the grain, tuber or fruit, and strengthens cell walls, which helps the plant withstand heat, drought and disease. Yet in 2023 Pakistan applied about 1 kg of potash (K_2O) for every 83 kg of nitrogen.

Made for the filling stage

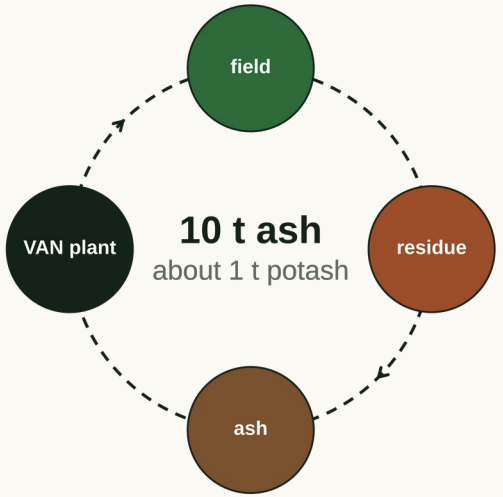
Vital Potash is fully water-soluble. It is applied in irrigation water, by fertigation or in flood irrigation, once the crop has stopped producing new leaves and has started to fill. Its 11% nitrogen gives the plant some nitrogen during filling, too little to start a new burst of leaf growth.

Boron coated onto the granule, in 2 forms

Since 2014 we have coated boron onto the potash granule on our own line, developed and built locally: sodium borate for fast action and calcium borate for slow release.

We do not state a boron percentage or a foliar rate. We do not claim an effect on yield, disease or plant immunity, or that 2 applications supply all the potash a crop needs in a season.

We recover potash from ash and use it in place of imported potash.



10 t

of crop residue ash yields about 1 t of potash

About 20%

of the potash we used in 2026 to date came from crop residue ash

2027

planned start of a larger potash recovery line

Silicon

recovery from the same ash is still in research and development, at the final stage; recovered silicon is not yet in any product.

V-Compost

is made only from plant residue, which is screened before use. It adds organic matter to soils that have less than 1% organic matter.

Recovering potash from the ash returns to the land the minerals that were removed from the field with the crop residue.

What we are developing.

Potash recovered in Pakistan

Potash recovery from crop residue ash will move to a larger line, planned for 2027.

Biologicals: mycorrhizae and phosphate-solubilising bacteria

We are developing biological products for Pakistani soil and climate: phosphate-solubilising bacteria from strains isolated in Pakistan, and, with a large European biological company, 2 mycorrhizal products to be made in Pakistan: a stand-alone mycorrhizal product, and a mycorrhizal-based phosphorus product that will be coated.

Mycorrhizal inoculant

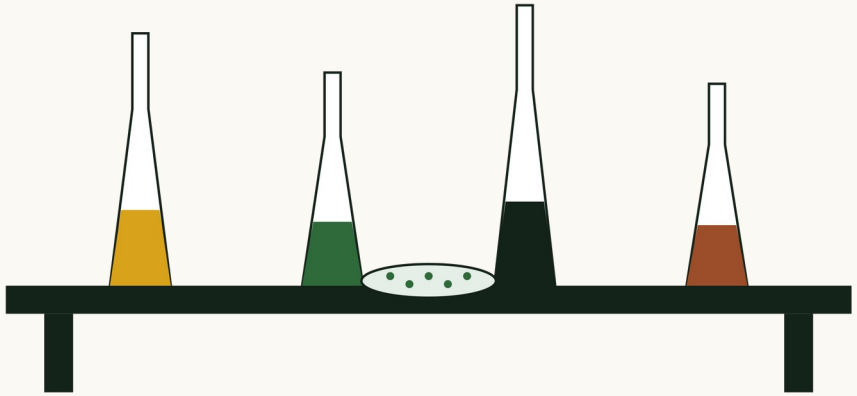
In development. Mycorrhizal fungi extend the reach of the roots, helping them take up the phosphorus that the bacteria release.

New coated and phosphate-efficiency products

Dual coated potash (K-50DC) is at concept stage, and dual release phosphate (P-DRC) at feasibility stage.

An NP granulation line

Planned for the plant site. Work has not started, and no date has been set.



The formulation library holds 27 formulations: 10 ready for your brand, 9 validated, 3 in field trials, 3 in development, 1 at concept stage and 1 at feasibility stage. Each is listed on van.com.pk with its code and stage.

TALK TO VAN

Each enquiry has its own address.



I grow

kisan@van.com.pk

A crop plan, product rates, a dealer near you.

Made under my name

partner@van.com.pk

White labelling, joint product development, toll manufacturing.

Everything else

info@van.com.pk

Supply, distribution, the laboratory, a plant visit.

Muhammad Ali
Chief Commercial Officer
muhammad.ali@van.com.pk

Fahim Asghar
Lead, Plant and Product
fahim@van.com.pk

Bilal Ahmed Khan
Chief Regulatory Officer
bilal@van.com.pk

Muhammad Irfan
Regional Business Head, Sindh
irfan@van.com.pk

WhatsApp +92 300 5003041 · Landline +92 42 35762215 · www.van.com.pk

Plant: 49-Km Multan Road, Phool Nagar Bypass, Lahore · Head office: 605 Al-Hafeez Heights, Gulberg III, Lahore 54660